

Campus Marketplace

Sprint 1 — Product Backlog & Delivery Plan

Software Design 2026 | Prepared: 30 March 2026 | Backend: Firebase

1. Sprint 1 Overview

Sprint 1 establishes the foundation of the Campus Marketplace platform. The sprint focuses on two core epics — user identity and authentication (E1) and item listing management (E2). Completing these epics unlocks all buyer and transaction features planned for Sprint 2.

1.1 Sprint 1 Story Summary

Epic	User Story	Sprint	Priority	Story Points
E1: User Verification & Roles	US1 – Student Registration & Login	Sprint 1	High	5
E1: User Verification & Roles	US2 – Role Assignment	Sprint 1	High	3
E2: Item Listings	US3 – Create an Item Listing	Sprint 1	High	5
E2: Item Listings	US4 – Edit or Remove a Listing	Sprint 1	Medium	3
E2: Item Listings	US5 – Mark Listing Type	Sprint 1	Medium	2

Sprint goal: Deliver a working registration, login, and role system backed by Firebase Auth, together with full listing creation, editing, and type-flagging capability.

2. Epic E1 — User Verification & Roles

This epic ensures every user on the platform is properly authenticated and assigned a role that determines their access. All tasks in E1 must be completed before Sprint 2 work can begin.

US1 — Student Registration & Login [High Priority]

As a student, I want to register and log in using a third-party identity provider so that my identity is verified and my account is secure.

Tasks:

Task	Assignee	Status
Research & choose identity provider (Auth0 / Firebase Auth / Google OAuth)	TBD	To Do
Configure identity provider credentials and app settings	TBD	To Do
Build login and registration page UI	TBD	To Do
Integrate frontend auth flow with chosen provider	TBD	To Do
Store user profile in Firestore on first login	TBD	To Do
Implement route guards — redirect unauthenticated users to login	TBD	To Do

Acceptance Tests (Given / When / Then):

Given	When	Then
A new student visits the site	They click Sign Up and complete the identity provider flow	Their account is created and they are redirected to the home page as a Student role
A registered student is logged out	They enter valid credentials	They are authenticated and land on their dashboard
An unauthenticated user tries to access a protected page	They navigate directly via URL	They are redirected to the login page

US2 — Role Assignment [High Priority]

As an admin, I want to assign roles (Student, Trade Facility Staff, Admin) to users so that each user only has access to features appropriate to their role.

Tasks:

Task	Assignee	Status
Define roles in Firestore user schema	TBD	To Do
Implement role-based access control (RBAC) using Firebase custom claims	TBD	To Do
Build admin UI to view users and assign or change roles	TBD	To Do
Write tests confirming Students cannot access Staff or Admin pages	TBD	To Do

Acceptance Tests (Given / When / Then):

Given	When	Then
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A logged-in admin opens the user management panel	They assign a user the Trade Facility Staff role and save	That user gains staff feature access and loses student-only access
A student user is authenticated	They attempt to access the admin dashboard URL directly	They receive an Access Denied response

3. Epic E2 — Item Listings

This epic covers the full lifecycle of a marketplace listing — creation, editing, deletion, and listing type designation. These stories enable the seller experience and provide the data foundation for buyer-facing features in Sprint 2.

US3 — Create an Item Listing [High Priority]

As a student seller, I want to create an item listing with photos, description, category, condition rating, and price so that other students can discover and purchase or trade my item.

Tasks:

Task	Assignee	Status
Design listing creation form UI	TBD	To Do
Implement photo upload (min. 1, max. 5) using Firebase Storage	TBD	To Do
Build category dropdown (Textbooks, Electronics, Furniture, Clothing, Other)	TBD	To Do
Build condition rating selector (New, Like New, Good, Fair, Poor)	TBD	To Do
Add listing type toggle: For Sale / For Trade / Either	TBD	To Do
Save listing document to Firestore linked to seller UID	TBD	To Do
Show success confirmation and redirect to new listing page	TBD	To Do

Acceptance Tests (Given / When / Then):

Given	When	Then
A logged-in student completes all required listing fields	They submit the form	A new listing is created and visible on the marketplace
A student creates a listing and uploads a photo	The form is submitted	The photo is stored in Firebase Storage and displayed on the listing page
A student submitting a listing leaves a required field empty	They click Submit	The form shows a validation error and does not submit

US4 — Edit or Remove a Listing [Medium Priority]

As a student seller, I want to edit or delete my own listings so that I can keep them accurate or remove items that are no longer available.

Tasks:

Task	Assignee	Status
Show Edit and Delete buttons only to the listing owner	TBD	To Do
Build pre-populated edit form from Firestore data	TBD	To Do
Implement delete with confirmation dialog	TBD	To Do
Add Firestore security rules preventing non-owners from editing or deleting	TBD	To Do

Acceptance Tests (Given / When / Then):

Given	When	Then
A logged-in seller views their own listing	They click Edit, update the price, and save	The listing reflects the new price immediately
A logged-in seller views their own listing	They click Delete and confirm	The listing is removed from the marketplace
A student who did not create a listing	They attempt to access the edit URL directly	They are denied access with an error

US5 — Mark Listing Type [Medium Priority]

As a student seller, I want to mark my listing as for sale, for trade, or either so that buyers know what kind of exchange I am open to.

Tasks:

Task	Assignee	Status
Add listing type field to Firestore listing schema	TBD	To Do
Display listing type badge prominently on listing card and detail page	TBD	To Do
Allow filtering by listing type on the browse page	TBD	To Do

Acceptance Tests (Given / When / Then):

Given	When	Then
A seller creates a listing and selects For Trade	The listing is published	A For Trade badge is visible on the listing card and detail page
A buyer browses listings and filters by For Sale	The filter is applied	Only listings marked For Sale or Either are shown

4. Dependencies & Technical Notes

- Firebase Auth must be configured before any frontend login work can begin (US1, Task 1-2).
- Firestore user schema (US2) must be agreed upon before US1's profile-storage task proceeds.
- Firebase Storage bucket permissions must be set up before photo upload (US3) can be tested end-to-end.
- Firestore security rules for listings (US4) depend on the role schema defined in US2.
- Frontend framework has not yet been finalised — the team should confirm this at the Sprint 1 kick-off.

5. Definition of Done

- All acceptance tests pass with no failing scenarios.
- Code is reviewed and merged to the main branch via pull request.
- Firestore security rules are deployed and tested in the Firebase console.
- UI components are responsive and tested on mobile and desktop viewports.
- No critical or high-severity bugs remain open at sprint close.